The 4th National Risk Management Competition for Undergraduate and Graduate Students
Samsung Fire & Marine Insurance × POSTECH × SNU

# AI-based Malicious Comment Legal Risk Insurance Service

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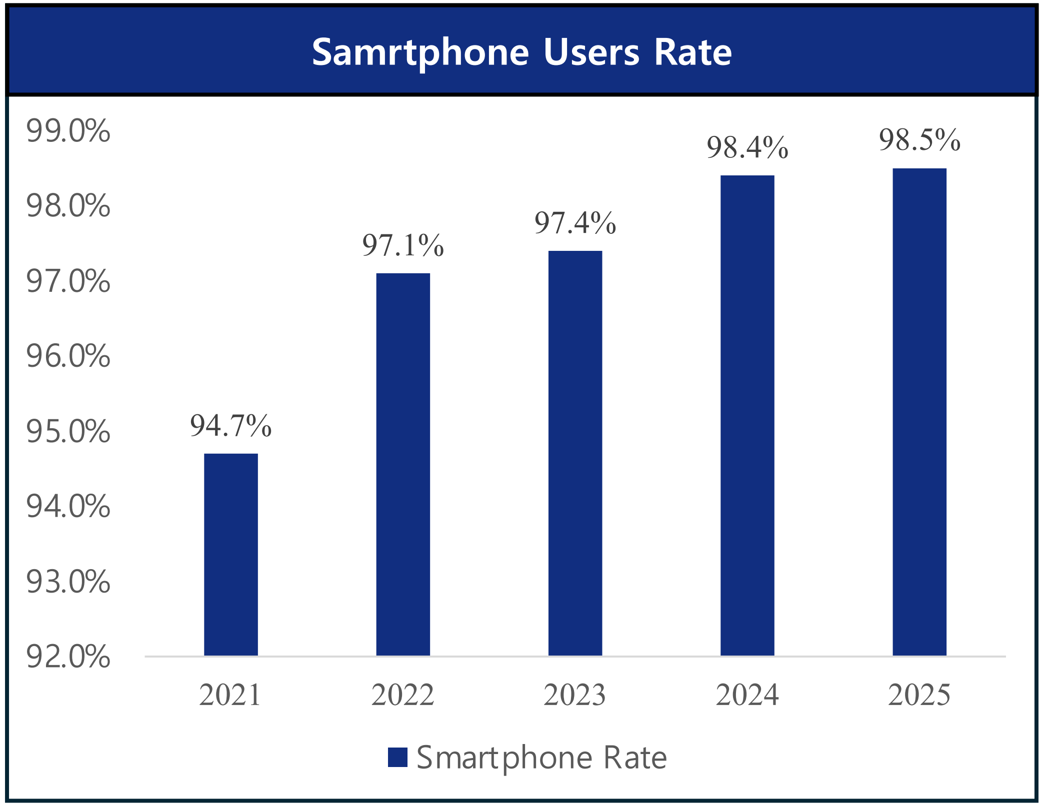
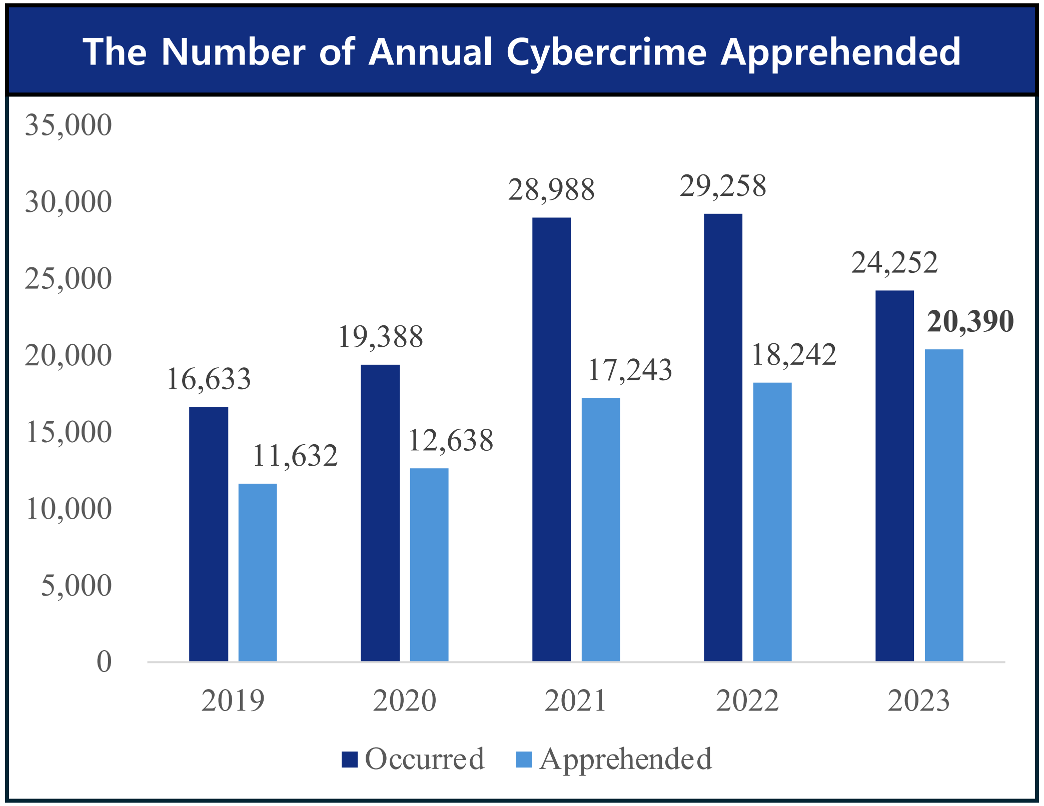
# I. Problem Definition

As smartphone usage and online contents have been increased,
the number of cybercrime apprehensions has doubled over the past five years.




* Source: GallupReport 20250707_Smartphone by Gallup (2025.07.03)



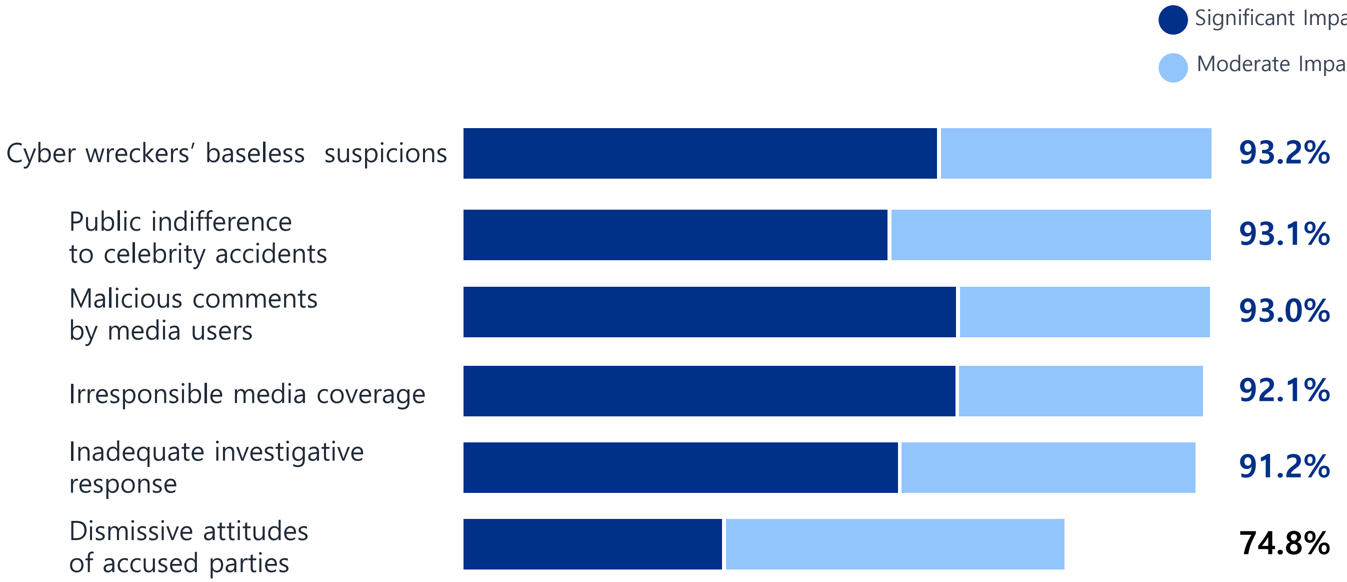

* Source: Annual Statistics on Cybercrime by Korea National Police Agency (2023.12.31)

**As cybercrime continues to rise, small influencers and independent entertainers increasingly need stronger protection against online harm.**

# I. Problem Definition

## The Fatal Impact of Online Malice

### < Perceived Impact of Factors on Celebrity Suicide Cases >




Korea Press Foundation Media Research Center | Online Survey, Feb 2024 | N=1,000 (Unit: %)

- Over 90% of respondents agree that online hate, cyber wreckers and irresponsible media are the leading causes of celebrity suicides
- A vast majority of the public views cyberbullying and media sensationalism as critical factors driving tragic outcomes.

## Structural Barriers for Victims

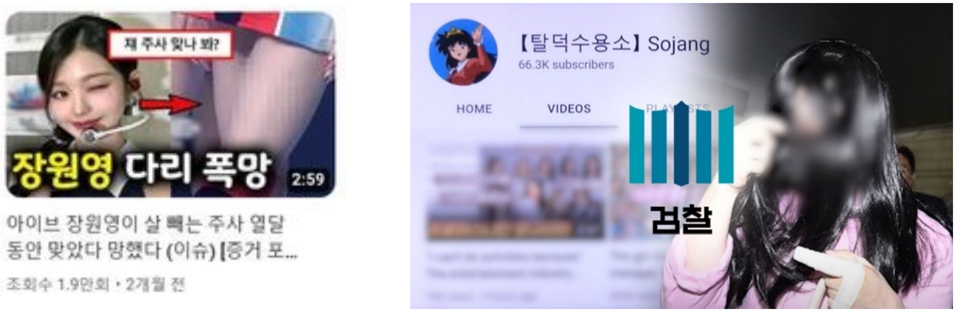



Figure 1. Cyber-Defamation and Criminal Case of YouTube Channel 'Sojang'"

**International Jurisdiction Barrier**

Overseas platforms (Google/YouTube HQ) systematically refuse disclosure, citing US 1st Amendment (Freedom of Speech) as a legal shield against foreign mandates.

**Prohibitively Complex U.S. Discovery**

Investigation stalls domestic side; requires independent petition to U.S. Feder al Courts. Legal fees often exceed hundreds of millions of KRW per case.

**The 'Sojang' Case: A Telling Precedent**

The 'Sojang' case shows success is reserved for corporate capital (Starship Ent .). For individuals, independent prosecution is practically impossible.

Source: YTN Radio "Case X-File" (Nov 10, 2025) | Atty. Jihyun Park, Royal Law Firm

**Legal Response to Malicious Comments Is Too Complex for Victims**



Sources: MBC News (Sep 3, 2024), "Prosecutors additionally indict 'Sojang' operator for defaming celebrities, Newsis (Aug 2, 2023), "Malicious Cyber-Wreckers Still Rampant Despite 'Sojang' Lawsuit"

# II. Current Limitations

Legal action is **complex, costly, and time-consuming** . While celebrities can rely on agency legal teams, individual creators often lack the resources to respond, leaving many cybercrime cases unreported.

**Current Procedure for Reporting Cyber Insults (by Goun Law Firm)**

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Collecting Evidence** | Page Screenshot | → | Print (HTML or PDF) | → | URL, Link Save | → | Internet Archive Generate | → | Meta Data Check |
| **Reporting** | Report Platform To save record | → | Visit Police With Evidence | → | Sue Accusation | → | Investigation Process | → | Claim for Damage |

When individuals handle this process alone, legal process can take months. Even after waiting up to seven months, cases are often dismissed due to insufficient evidence or unmet legal requirements, **causing wasted time and additional stress.**

**Youtuber A**

The cybercrime incident took place while the victim was playing Valoran.'

| (2025.03.04) | (2025.03.17) | (2025.03.20) |
|---|---|---|
| Sue Accusation | Investigation | **Non-referral decision** |

# II. Current Limitations: AS-IS / TO-BE

Make Cybercrime Reporting Process Easier for Reducing the burden on victims and practitioners through AI-driven automation

## AS-IS

- Victim faces malicious comment
- Manual evidence collection
- Hard to judge legal eligibility
- Complex and time-consuming process
- Cases often dismissed or left unreported

### CURRENT PROCESS BOTTLENECKS

1. High financial and procedural burden
2. Long waiting time for manual review
3. Heavy manual workload for practitioners

## TO-BE

- Victim uploads comment or evidence
- AI extracts and organizes evidence
- **AI classifies risk and legal relevance**
- Practitioners review prioritized cases
- Faster, easier, and consistent response

### FUTURE VALUE DELIVERY

1. Lower reporting burden for victims
2. Automated & accurate risk classification
3. Reduced practitioner manual workload

# III. AI-driven Comment Detection Service Flow

From evidence submission to automated risk assessment and smarter insurance pricing

### Evidence Submission

Users upload evidence through the reporing portal

### OCR & Text Extraction

AI extracts comments, author, timestamp.. Etc from PDFs, screenshots, and webpages.

### AI Risk Scoring

Eash comment is scored on 7 legal-risk indicators from a **evaluating model**

### High-Risk Comment Filtering

**AI prioritizes** high-risk comments, reducing the review burden for insurance practitioners and legal partners

### Case-Level Aggregation

Comments grouped at the post or video level, rather than treated as isolated items, to calculate overall case risk

### Risk Database Accumulation

Risk scores, settlement likelihood, expected legal costs, and recovery amounts are accumulated to improve pricing

Streamline cybercrime complaint handling through AI-powered evidence analysis and risk classification automatically

# III. AI-driven Comment Detection Service Flow: Scoring Modeling

## Risk Scoring Development Process

**STEP 1**

**Data Collection**

Using Youtube API to collect the comment data to train th e risk modelling

**Count: 185,618**

**STEP 2**

**AI Filtering**

AI Filters the whole comments to classify the suspicious revie w

**Count: 14,916**

**STEP 3**

**Manual Indexing**

Our team manually indexed a nd labeled the comments bas ed on seven legal-risk factors

**Count: 885**

**STEP 4**

**Scoring Model**

Trained the scoring model on manually indexed comments across seven legal-risk factors and applied it to the full data set.

**Count: 185,618**

| Factors | 1 (Low) | 2 (Vague) | 3 (High) | Note |
| --- | --- | --- | --- | --- |
| Projanity Intensity | 263 | 394 | 228 | |
| Factual Allegation | 365 | 283 | 237 | |
| Sexual Expression | 531 | 108 | 246 | |
| Threat | 840 | 46 | 23 | Point 1 is majority |
| Repetition | 816 | 46 | 23 | Point 1 is majority |
| Identifiability | 151 | 218 | 516 | Most comments included the victim's name |
| Publicity | 0 | 0 | 885 | **Because the comments generated in Youtube, all satisfied the 'Publicity'** |

# III. AI-driven Comment Detection Service Flow

## Best Algorithms to be used in Models (5-fold CV)

| Facotrs | Algorithms | Macro F1 | Accuracy | MAE |
|---|---|---|---|---|
| Projanity Intensity | Multinominal_Navie Basiean | 0.4814 | 0.4947 | 0.6250 |
| | Complement_Navie Basiean | 0.4388 | 0.4542 | 0.7051 |
| | Log_Odds_OvR | 0.4741 | 0.4822 | 0.6443 |
| | Ridge_OvR | 0.5270 | 0.5377 | 0.5561 |
| | Nearest_Centroid | 0.5251 | 0.5320 | 0.5799 |
| Factual Allegation | Multinominal_Navie Basiean | 0.5545 | 0.5607 | 0.5218 |
| | Complement_Navie Basiean | 0.5453 | 0.5595 | 0.5467 |
| | Log_Odds_OvR | 0.5487 | 0.5561 | 0.5320 |
| | Ridge_OvR | 0.5767 | 0.5889 | 0.4811 |
| | Nearest_Centroid | 0.5439 | 0.5471 | 0.5432 |
| Sexual Expression | Multinominal_Navie Basiean | 0.5502 | 0.6564 | 0.4691 |
| | Complement_Navie Basiean | 0.5308 | 0.5761 | 0.5369 |
| | Log_Odds_OvR | 0.5533 | 0.6597 | 0.4590 |
| | Ridge_OvR | 0.5928 | 0.7457 | 0.3549 |
| | Nearest_Centroid | 0.5502 | 0.6734 | 0.4577 |

| Facotrs | Algorithms | Macro F1 | Accuracy | MAE |
|---|---|---|---|---|
| Threat | Multinominal_Navie Basiean | 0.3804 | 0.8723 | 0.1887 |
| | Complement_Navie Basiean | 0.3532 | 0.7006 | 0.4452 |
| | Log_Odds_OvR | 0.3780 | 0.8655 | 0.1966 |
| | Ridge_OvR | 0.3839 | 0.9413 | 0.0881 |
| | Nearest_Centroid | 0.4033 | 0.8475 | 0.2260 |
| Repetition | Multinominal_Navie Basiean | 0.3444 | 0.8022 | 0.2656 |
| | Complement_Navie Basiean | 0.2409 | 0.4055 | 0.8861 |
| | Log_Odds_OvR | 0.3419 | 0.7671 | 0.3120 |
| | Ridge_OvR | 0.3911 | 0.9085 | 0.1197 |
| | Nearest_Centroid | 0.3398 | 0.7513 | 0.3549 |
| Identifiability | Multinominal_Navie Basiean | 0.5038 | 0.6067 | 0.5132 |
| | Complement_Navie Basiean | 0.5044 | 0.5797 | 0.5707 |
| | Log_Odds_OvR | 0.5135 | 0.6158 | 0.5052 |
| | Ridge_OvR | 0.5324 | 0.6452 | 0.4679 |
| | Nearest_Centroid | 0.5335 | 0.6350 | 0.4928 |

*Calculation are all based on Youtube comments. So, We defined that all comments satisfied 'Publicity' factor

## Crime-Specific Combination Rules and Case-Level Aggregation

### 1. Score-to-Probability Mapping

To convert each comment's 7 indicator scores from **1–3** into probabilities between **0 and 1**, we applied a monotonic mapping rule.

| Factor Score | Probability of Satisfying Legal Requirements (Case 1) | Probability of Satisfying Legal Requirements (Case 1) | Meaning |
|---|---|---|---|
| 1 | 10% | 20% | Mostly not |
| 2 | 50% | 50% | Need Evaluation by people |
| 3 | 90% | 80% | Mostly Satisfy |

Because specific success rates for each legal requirement were not available, we assumed two scenarios for insurance pricing

### 2. Crime-Specific Combination Rules

For 4 major offenses under Korean criminal law, we represented the legal requirements as multiplicative combinations of the seven indicators. This reflects the legal principle that all required elements must be satisfied for an offense to be established.

| Offense | Combination Rule | Legal Basis |
|---|---|---|
| Insult | P(Profanity Intensity) x P(Identifiability) x P(Publicity) | Criminial Act, Article 311 |
| Defamation | P(Factual Allegation) x P(Identifiability) x P(Publicity) | Criminal Act, Article 307 |
| Intimidation | P(Threat) x P(Identifiability) | Criminal Act, Article 283 |
| Obscenity via Communication Media | P(Sexual Explicitness) x P(Identifiability) x P(Publicity) | Act on Special Cases Concerning the Punishment of Sexual Crimes, Article 13 |

# III. AI-driven Comment Detection Service Flow

## Probability

| Offense | Accusable Probability(P_i) | Unit Legal Cost(B_i) |
|---|---|---|
| Insult | 0.2245 | KRW 500,000 |
| Defamation | 0.1859 | KRW 1,000,000 |
| Intimidation | 0.1153 | KRW 1,000,000 |
| Obscenity via Communication Media | 0.1260 | KRW 1,500,000 |

## Premium

Assumption: 1 Incident = 1 Comment

$$\text{Total Expected Cost} = \sum P_i \cdot B_i$$

$$\text{Total Gross Cost} = \text{Total Expected Cost} \times 1.4$$

TOTAL VALID INCIDENTS

**1,369**

ANALYZED COMMENTS

**191,388**

TOTAL EXPECTED COST (KRW)

**115,279,938,000**

FINAL TOTAL GROSS COST (KRW)

**161,391,913,200**

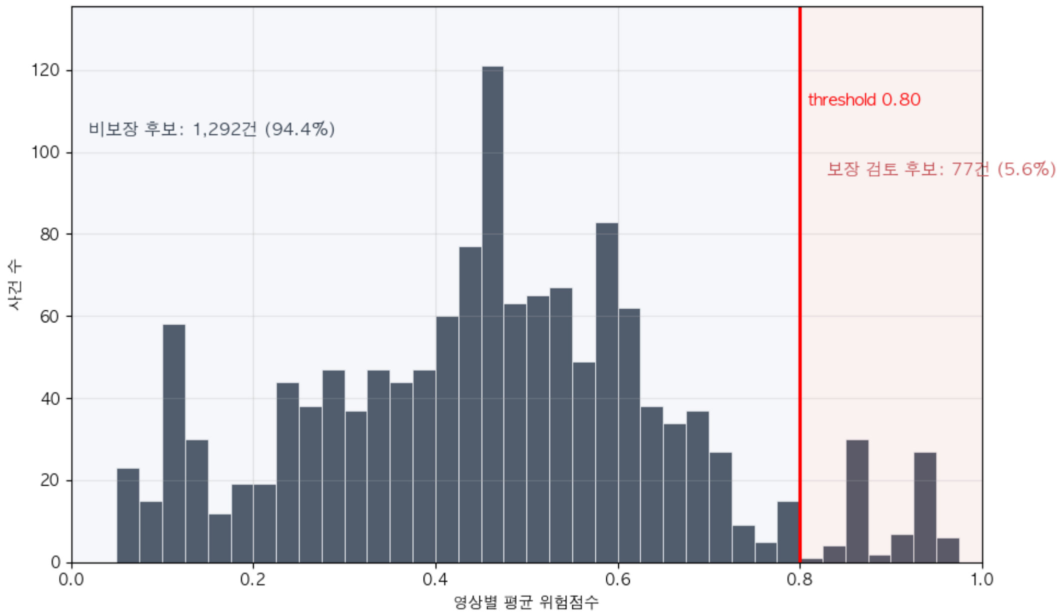
**Key Point**: Method captures individual comment-level legal risks, but **may overestimate the cost when multiple comments** belong to the same case

## Premium

To make insurance pricing more realistic and manageable, we **defined each incident as a single video and calibrated a threshold-based classification method** to **identify the highest-risk 4–5% of cases, reflecting typical insurance coverage rates.**

$$Risk_j = 1 - (1 - P_{Insult})(1 - P_{Defamation})(1 - P_{intimidation})(1 - P_{Obscenity}) \text{ (Per Video)}$$

$$AvgRisk_j = \frac{1}{|C_i|} \sum Risk_j \ (Per\ Video)$$




**Threshold was set at 0.8**

So if AvgRisk is greater than or equal to 0.8…

① Comments risks are averaged at the video level to reflect the actual claim unit

② Videos with AvgRisk $\geq$ 0.8 are classified as potential covered events

③ As a result, 77 videos (5.6%) are selected as coverage candidates

This approach aligns with the claim process, but severe individual comments may be diluted by averaging

# III. AI-driven Comment Detection Service Flow

## Individual Gross Cost Comparison

$$Benefit_i = Covered_i \times 1{,}000{,}000$$
$$Gross\ Premium_i = Benefit_i \times 1.4$$
$$Total\ Gross\ Premium = \Sigma\, Gross\ Premium_i$$

* KRW 1 million assumed per covered claim, with a 40% insurer premium margin

$$Covered_i = \begin{cases} 1 & AvgRisk \geq 0.8 \\ 0 & AvgRisk < 0.8 \end{cases}$$

**[Before]**

**Individual Gross Cost (1 Accident = 1 Comment)**

| Insured | Accident | Gross Cost |
|---|---|---|
| S | 204 | 47,111,930,000 |
| K | 356 | 32,068,510,000 |
| P | 25 | 2,538,250,000 |
| Others | 784 | 79,673,220,000 |

**[After]**

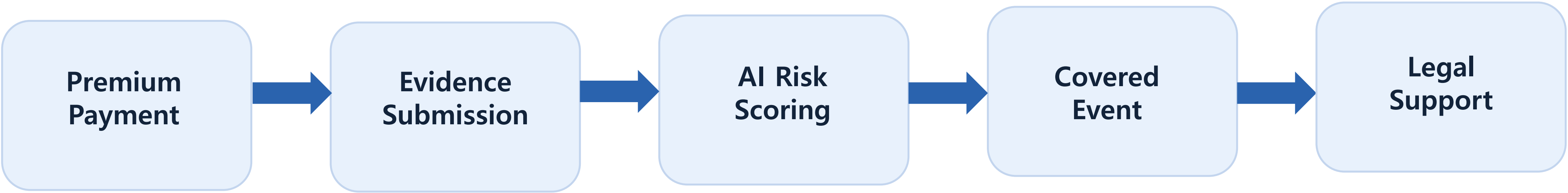
**Individual Gross Cost (1 Accident = 1 Video)**

| Insured | Accident | Gross Cost |
|---|---|---|
| S | 9 | 12,600,000 |
| K | 23 | 32,200,000 |
| P | 1 | 1,400,000 |
| Others | 44 | 61,600,000 |

**After redefining accidents and applying the threshold,
the estimated annual insurance cost decreased significantly**

# IV. Insurance Product Design

AI-powered Legal Expense Insurance product that helps individual creators detect file and respond to malicious comments more efficiently. The revenue model will be generated as follows, and AI will be used in this process.




This product supports legal response costs when the insured suffers online harm and may later recover part of the costs from the perpetrator.

# IV. Insurance Product Design

## Service Flow: Automated Process Engine Architecture

### EVIDENCE SUBMISSION 01

- User uploads PDF or image file containing comments an d metadata via the insurer portal.
- System initializes case ID and logs the user identity time stamp.

### TEXT & META EXTRACTION 02

- Executes OCR engine processing to extract raw text cont ent from images/PDFs.
- Simultaneously identifies and separates structural metad ata (URLs, timestamps, account IDs).

### AI RISK SCORING 03

- Parses raw text data into the Legal Risk Evaluation Engin e.
- Evaluates and scores comments based on 7 statutory fac tors (Insult, Defamation, Threat, Repetition, Public Expos ure, Identifiability, Obscenity).

### HIGH-RISK FILTERING 04

- Applies probabilistic threshold rules to classify content.
- Filters out low-risk data and isolates the highly actionabl e comment pool to minimize manual legal review overh ead.

### CASE-LEVEL AGGREGATION 05

- Aggregates filtered comments at the Post or Video level.
- Calculates total expected legal costs, case priority, and s ettlement success metrics per incident.

### DATABASE ACCUMULATION 06

- Archives processed incident data into the master databa se.
- Feeds accumulated risk logs back into the Actuarial Prici ng Engine to optimize underwriting and variable premiu ms.

**AI-powered legal expense insurance product that helps individual creators detect, file, and respond to malicious comments more efficiently**

# V. Implementation Strategy

## Micro-insurance Product Roadmap
## : From Individual Pilot to B2B Scale-up

The service starts as an entry-level pilot for solo media creators, then evolves into a sophisticated data-driven underwriting model integrated with legal networks, eventually expanding into comprehensive B2B enterprise products for MCNs and platforms.

### Micro-insurance Development Roadmap

| Evolution Phase | Phase 1. Micro Insurance Pilot | Phase 2. Risk Scoring Enhancement | Phase 3. Integration | Phase 4. Scale-up |
|---|---|---|---|---|
| Target & Definition | **Solo Creators & Streamers** | **Data-Driven Risk Assessment** | **Automated Litigation Solutions** | **B2B Enterprise & MCNs** |
| Strategic Actions & Key Deliverables | • Launch low-premium entry models for solo creators and influencers<br>• Establish monthly subscription-based premium systems<br>• Include comment analysis and legal counseling support sessions | • Develop scoring variables: follower count, avg. views, genre and exposure to controversy<br>• Define risk weightage by individual profile and content categories<br>• Implement video-level incident threshold adjustment modules | • Establish real-time connectivity with partner law firms and attorneys<br>• Provide automated templates for legal complaints and filings<br>• Build automated evidence package generation systems | • Refine underwriting based on accumulated risk data, settlement fees, and legal costs<br>• Full expansion to B2B products tailored for MCNs and major digital platforms |

# VI. Expected Social Impact

The Insurance Service eradicates malicious comments and cyber-wreckers by **empowering victims and opening new markets for insurers**.

## Victim Protection

### IMPROVED ACCESSIBILITY

- **Agency-Free Redress:** Individual influencer s can initiate legal action without complex agency representation.
- **Simplified Procedures:** Victims no longer need to navigate intricate complaint syste ms alone.

### PROACTIVE ENFORCEMENT

- **Cyber-Wrecker Deterrence:** Pursuit of cas es beyond personal channels to include m alicious "fake news" content.
- **Beyond Reporting:** More effective than rel ying on slow and often limited platform re porting systems.

## Insurer Value

### NEW MARKET MODEL

- **Category Leadership:** Establishing a specia lized market centered on online reputation and digital risk.
- **Data Monetization:** Utilizing accumulated legal data to drive product innovation and differentiation.

### STRATEGIC MANAGEMENT

- **Precise Underwriting:** Leveraging incident frequency and settlement data to enhance risk assessment.
- **Loss-Ratio Control:** Deterrence-led model s reduce systemic legal payouts over the lo ng term.

## Digital Ecosystem

### MALICIOUS DETERRENCE

- **Financial Consequences:** Awareness of set tlements, damages, and legal costs reduces commenter malice.
- **AI-Driven Detection:** Insurer-led legal resp onses create a scalable wall against cyber-wreckers.

### ENVIRONMENT IMPROVEMENT

- **Cultural Shift:** Reduced tolerance for toxicity strengthens digital community accountability.
- **Faster Resolution:** Rapid legal responses l ead to a healthier and safer online commu nication environment.